

MERWAN BIREM

AI Innovation Leader | Computer Vision, Industrial AI Systems & Autonomous Perception

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PROFILE

AI Innovation Leader with 10+ years of experience in computer vision, robotics, and industrial AI systems. Proven track record of leading multi-million euro R&D programs (4M+) and delivering applied AI solutions from research to industrial deployment. Specialized in perception systems, AI-driven quality inspection for complex manufacturing environments, and visual SLAM. Expanding expertise in AI systems engineering, MLOps, and workflow automation to bridge advanced research with scalable and reliable real-world deployment.

EXPERIENCE

Project Leader & Senior Research Engineer – AI Innovation Flanders Make

Apr 2022 – Present Leuven, Belgium

- Lead large-scale AI R&D programs at national and EU level, including the 4M ALARMM SBO project involving 4 research groups and 7 industrial partners.
- Represent the research group in national and European collaborations, contributing to strategic positioning and long-term research roadmaps.
- Actively involved in business development activities: identification of new research and innovation opportunities, partner engagement, and co-definition of future project proposals.
- Manage and coordinate cross-functional teams (3–12 members) across academia, industry, and public research stakeholders.
- Drive development of advanced AI systems in computer vision, visual SLAM, and sensor fusion for industrial environments.
- Achieved 4x technology transfer from research to industry in the most recent completed project (2025), enabling follow-up innovation initiatives.
- Oversee full project lifecycle: opportunity scouting, ideation, consortium building, proposal writing, technical execution, stakeholder alignment, and technology transfer.
- Mentor engineers and contribute to internal capability building, as well as strategic research and innovation roadmap definition.

Research Engineer – Computer Vision & AI Flanders Make

Jan 2018 – Mar 2022 Leuven, Belgium

- Designed AI-based vision systems for industrial inspection and automation.
- Applied deep learning and self-supervised learning for real-time perception systems.
- Contributed to EU-funded project proposals and technical work packages.

Research Engineer – Intelligent Vehicles Flanders’ DRIVE

May 2015 – Dec 2017 Lommel, Belgium

PhD Researcher – Embedded Vision for Robotics Institut Pascal

2011 – 2015 Clermont-Ferrand, France

CORE COMPETENCIES

AI & Perception Systems

- Computer Vision (2D/3D)
- Visual SLAM
- Sensor Fusion
- Deep Learning
- Pose Estimation
- Object Detection
- Quality Monitoring
- AR/VR for Industry

Tools Currently using & used before

- Python
- OpenCV
- Open3D
- ROS2
- Git
- Docker & Containerization
- CI/CD
- MLOPS
- PyTorch
- C/C++
- VHDL
- MATLAB
- JAVA

Emerging Focus - Learning

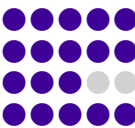
- AI Workflow Automation (n8n)
- AI Agents
- LLM Applications
- MCP

EDUCATION

- PhD in Electronics & System Architecture, Blaise Pascal University, (2011-2015), France
- MSc in Robotics & Multisensory Perception, Blaise Pascal University, (2010-2011), France
- Engineering Degree in Electronics, National Polytechnic School, (2005-2010), Algeria

LANGUAGES

English
French
Dutch
Arabic



PUBLICATIONS

 Full list of publications available on Google Scholar.

Journal Articles

- Weckx, Sam et al. (2022). "A cloud-based digital twin for monitoring of an adaptive clamping mechanism used for high performance composite machining". In: *Procedia Computer Science* 200, pp. 227–236.
- Birem, Merwan, Richard Kleihorst, and Norddin El-Ghouti (2017). "Visual odometry based on the Fourier transform using a monocular ground-facing camera". In: *Journal of Real-Time Image Processing*, pp. 1–10.
- Birem, Merwan and François Berry (2014). "DreamCam: A modular FPGA-based smart camera architecture". In: *Journal of Systems Architecture* 60.6, pp. 519–527.

Conference Proceedings

- Birem, M, C Domken, and A Bey-Temsamani (2024). "CAD2GraspMonitor: Vision-based Robotic Grasp Monitoring for Industrial Kitting Application". In: p. 129.
- Ruosch, C, M Birem, and A Bey-Temsamani (2024). "Bench Marking of Industrial Object Pose Estimation Algorithm Based on Vision & Deep Learning". In: p. 177.
- Solmaz, Serkan et al. (2024). "Capturing the system requirements for adopting Industry 5.0 in quality inspection: an evidence-based approach". In: vol. 128. Elsevier, pp. 369–374.
- André, Hugo et al. (2021). "Using a smartphone camera to analyse rotating and vibrating systems: Feedback on the SURVISHNO 2019 contest". In: vol. 154. Academic Press, p. 107553.
- Gors, Dorothy, Merwan Birem, et al. (2021). "An adaptable framework to provide AR-based work instructions and assembly state tracking using an ISA-95 ontology". In: vol. 104. Elsevier, pp. 714–719.
- Zogopoulos, Vasilios et al. (2021). "Image-based state tracking in augmented reality supported assembly operations". In: vol. 104. Elsevier, pp. 1113–1118.
- Dwek, Nathan et al. (2019). "Improving the accuracy and robustness of ultra-wideband localization through sensor fusion and outlier detection". In: vol. 5. 1. IEEE, pp. 32–39.
- Kikken, Edward et al. (2019). "A model-based approach to enhance performance in high-end laser-positioning systems". In: *2019 20th International Conference on Research and Education in Mechatronics (REM)*. IEEE, pp. 1–6.
- Gors, Dorothy, Robbert Hofman, et al. (2017). "New Object Tracker Based On Adaptive Intensity Models of Object and Its Surroundings". In:
- Shopovska, Ivana et al. (2017). "A hybrid fusion based frontal-lateral collaborative pedestrian detection and tracking". In: *2017 IEEE 20th International Conference on Intelligent Transportation Systems (ITSC)*. IEEE, pp. 1–6.
- Birem, Merwan, Jean-Charles Quinton, et al. (2014). "SAIL-MAP: Loop-closure detection using saliency-based features". In: *2014 IEEE/RSJ International Conference on Intelligent Robots and Systems*. IEEE, pp. 4543–4548.
- Quétard, Boris et al. (2014). "Saliency in the fog: application to real and 2.5 D pictures". In: *TRA2014 Transport Research Arena 2014*.

PHILOSOPHY

"Life is like riding a bicycle. To keep your balance, you must keep moving."
– Albert Einstein

KEY PROJECTS

ALARMM SBO – Visual SLAM for Large Environments - [2025 – Present]

- Leading 4M multi-partner project on low-cost, accurate and robust vision SLAM systems
- Focus on robust perception in complex industrial environments
-  Latest project update: LinkedIn post - dissemination (April-2026)

QUALMA ICON – AI-Augmented Quality Assurance - [2022–2025]

- Led development of modular AI framework for large-scale manual assembly inspection
- Integrated computer vision (detection, pose estimation, tracking) with AR guidance
- Improved inspection accuracy and reduced cycle time in industrial scenarios
-  Demo video: QUALMA ICON Industrial YouTube Project Demonstration

FROGS ICON CAD2GraspMonitor – Robotic Grasp Validation - [2021–2023]

- Developed vision-based system for post-grasp validation in robotic kitting
- Combined YOLO-based detection, ICP pose estimation, and sensor fusion
- Ensured high reliability in industrial pick-and-place operations